

Nikhil Prabhu

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EDUCATION

University of California, San Diego (UCSD)
B.S. in Computer Science | Cumulative GPA: 3.67

San Diego, CA
Expected Graduation: June 2028

SKILLS

Languages: Python, C++, TypeScript, JavaScript

Robotics Infra: ROS 2, MuJoCo, SLAM, visual odometry, embedded systems, real-time streaming pipelines

Backend: FastAPI, Node.js, Express, WebSockets, SQLite, Supabase, Docker, Kubernetes, Linux

ML + Tooling: PyTorch, PPO/MAPPO, OpenCV, YOLOX, Git, CI workflows, MCP tool servers

EXPERIENCE

Desktop Agent Developer & Founder — UCSD Agent Development (Clue2)

San Diego, CA

Mentored by Prof. Ramamohan Paturi

Jan 2025 – Present

- Architected a cross-platform Electron + TypeScript desktop agent with real-time meeting transcription, WebSocket streaming, dual-channel capture, buffering, and fault-tolerant session management.
- Built an LLM workflow engine across Gmail, Calendar, Docs, Sheets, Slack, Notion, and 7+ services; scaled beta to 40+ users with YC-startup adoption.

Undergraduate Research Assistant — AI for Circuit Invention, UCSD

San Diego, CA

Research Assistant (with Prof. Prasad Gudem & Samvid AI)

March 2026 – Present

- Built infrastructure connecting LLM APIs to Cadence simulation tools, automating generate-simulate-evaluate loops and parsing Spectre output into structured benchmarks for RL design-space search.

Lead Researcher — Foot Drop Ankle-Foot Orthosis (Georgia Tech mmWave Lab)

Atlanta, GA

IEEE ICHCI 2024 Presenter

May 2024 – Nov 2024

- Built the embedded sensing + actuation stack for a 3D-printed EMG-driven orthosis ($R = 0.968$ correlation, 23% gait precision gain); first-author paper at IEEE ICHCI 2024.

ROBOTICS PROJECTS

SODIUM — In-Home Care Robot

1st Overall, DiamondHacks 2026

April 2026

- Built a ROS 2 robot stack with wake-word detection, NL task routing, medication reminders, and crisis detection — embedded on-device software with a cloud caregiver control loop.
- Shipped a lightweight Svelte 5 / Bun frontend powering both the in-home robot UI and the operator dashboard over real-time WebSocket sync.

DroneOS

1st Overall, Bow Capital × DS3 × SIC Hackathon (UCSD)

2026

- Built edge autonomy integrating SLAM, visual odometry, and YOLOX + Norfair perception for GPS-denied navigation; trained MAPPO swarm policies validated in simulation before deployment.

Battle Angel — MuJoCo Simulation Infrastructure

Google Gemini 3.5 Hackathon

2026

- Built a PPO training + evaluation harness over eight MuJoCo environments with a FastAPI backend, vectorized rollouts, GIF replay, and Gemini-generated disaster scene layouts.

Crucible Compute — Containerized Inference Control Plane

5th Place, Nozomio Hackathon

2026

- Built a GPU routing control plane across Docker, Kubernetes, Vultr, and Vast with spend-approval gates and an OpenAI-compatible inference gateway.
- Benchmarked cost/performance placements in real time and exposed MCP tools so agents can deploy workloads autonomously across cloud GPU providers.

PUBLICATIONS

- **Real-Time EMG Control for Assistive Devices** — IEEE ICHCI 2024.
- **Co-Heritability of Antibiotic Resistance in TB Evolution** — Bioinformatics & Biomedical Technology, 2025.